Verifier rubric: ten-day mathematical diagnostic

Version 1 · 17 October 2026 · Write this specification before generating items.

Purpose and limits

Identify what mathematics the learner knows, understands, can execute, recognizes and can combine. Use 10 daily 60-minute sessions. This is a criterion-referenced diagnostic prototype, not a normed psychometric test, IQ measure, AMC score predictor or validated ability scale. Estimated item times and difficulty labels require piloting. Geometry, trigonometry and probability are outside this agreed diagnostic scope; do not claim coverage of all high-school mathematics.

Modular arithmetic is limited to last digits, remainder cycles, sums/products of powers and exponent-position traps. Do not require abstract algebra, CRT, primitive roots or advanced modular theory. Vieta formulas belong in core algebra; Vieta jumping is an optional enrichment probe and cannot lower a baseline rating.

Claims → evidence → tasks

Every named skill needs four distinct probes: L1 knowledge/conditions, L2 direct use, L3 uncued recognition/application, L4 combination/extension. L3/L4 are design intentions inspired by contest problem solving, not calibrated AMC/AIME equivalences. At least one question per skill must ask for a reason, derivation, counterexample or explanation. A successful L1 recall question cannot establish application; a wrong calculation cannot by itself establish a missing concept.

Each item must specify its skill, level, prerequisites, exact task, domain/assumptions, expected answer, complete reasoning, item-specific scoring evidence, likely error patterns, neutral hint, estimated time and a follow-up rule. Student-facing wording must not reveal the method in L3/L4, skill labels or answers. All questions are original; no copied or falsely attributed contest questions.

Scored review criteria

Score 0–4: 0 missing, 1 major faults, 2 revisions needed, 3 meets specification, 4 supported by stronger evidence. Publish only if each criterion reaches 3, the weighted total reaches 85, and all gates pass. A spreadsheet score is not evidence of correctness; report the actual review method and limits.

Criterion	Weight	Evidence required
Topic and result coverage	20	Ledger maps algebra, inequalities, functions/functional equations, logic, summations, counting and elementary number theory to concrete items; all user examples included.

Criterion	Weight	Evidence required
Mathematical correctness	25	Verifier solves question-only tasks before inspecting proposed answers. Exact calculations/enumerations checked where possible. Written proofs inspected for missing cases, invalid implications, domain and endpoint errors. No unresolved disagreement.
Diagnostic separation	20	Items distinguish knowledge, execution, recognition and combination. Multiple observations support claims. Error codes are hypotheses with discriminating follow-ups, not automatic diagnoses.
Difficulty ladder and branching	15	Every skill has L1–L4; prerequisite failures lead to a parallel lower-level check; successful independent work can lead upward. Hard item failure does not erase secure lower-level evidence.
Timing and administration	10	Ten sessions, exactly 60 scheduled minutes each, fixed common core plus bounded adaptive time. Unreached, unanswered, incorrect, hinted and previously exposed are distinct. No hidden requirement to complete all reserve questions.
Clear language and usable artifacts	10	Simple precise prompts, no method hints in uncued tasks, correctly rendered notation, separate student/tutor books, clear question IDs, printable response/record sheets and navigable day files.

Mandatory release gates

1. Complete L1–L4 coverage ledger for every skill; no topic counted from a title alone.
2. Answer-guide completeness: every question and optional branch has an answer, reasoning, evidence anchors and follow-up.
3. Zero unresolved mathematical or task-ambiguity faults. Numerical verification must start from inputs, not compare constants copied from the answer key.
4. Each core appears once in the schedule; delayed confirmations use changed prompts. No question's solution may be printed in the student book.
5. Fit the 60-minute schedule with a hard stop, optional adaptive slots and incomplete-attempt handling. Estimates are labelled provisional.
6. No mastery from one item, no score percentile or latent-ability claim without calibration. Report skill/rung evidence and uncertainty.
7. Derivations and explanations have item-specific credit anchors. Accept equivalent correct methods; do not require the author's trick.

8. Any feedback/hint/answer exposure is logged. Delayed items after feedback measure retention/learning, not an uncontaminated baseline. Keep the Day-10 confirmation prompts unavailable to the learner until Day 10.
9. Distinguish unfamiliar terminology, prior exposure, slow retrieval, execution slip and conceptual confusion using follow-up evidence. Low confidence is not an error.
10. Review records identify who/what checked each claim. Same-author editorial review and answer-blind computation do not count as an independent human or model review. If independent review is absent, label release AUTHOR_REVIEWED_COMPUTATION_CHECKED, never independently certified.

Generator–verifier loop

For each day: generate question-only core and candidate branches → freeze the question inputs → solve/check without reading proposed solutions → compare against author solutions → inspect diagnostic quality and timing → repair any faults → rerun affected checks → write review findings → lock exact file hashes. Start the next day's release only after the current day is locked. Later changes invalidate dependent artifacts and require a new revision. Shared question bank drafting is allowed, but day releases remain sequential.

The verifier must record findings and concrete corrections, not merely PASS. Mechanical checks cover structure, algebraic/numeric oracles and file integrity. Author review covers language, proof completeness and interpretation. Independent validation is a separate status.

Administration and interpretation

Before beginning: explain that leaving an honest attempt is useful. No calculator, internet, AI, notes or formula sheet on the independent pass. Accommodations may change this; record them and do not compare times as equivalent. Use scratch paper. Record brief confidence 0 (guess), 1 (uncertain), 2 (fairly sure), 3 (sure), and familiarity: new/seen similar/seen exact. Timing is approximate active work, not a speed penalty.

Daily budget: 3 minutes setup; 30 minutes common core; 20 minutes for up to two targeted follow-ups; 7 minutes confidence/exposure/strategy log. Stop after 60 minutes. A student may leave core questions unreached; record that status rather than zeros. Continue within the adaptive budget only; no homework to make the diagnostic appear complete.

Prefer delaying worked-answer release until Day 10. Give neutral process acknowledgement rather than teaching between baseline sessions. If feedback or outside study occurs, log it. Day 10 distinguishes fresh confirmation from learning/retention as appropriate.

Record each item using outcome {independent-correct, partial, attempted-incorrect, not-attempted, not-reached, hinted-correct, hinted-incorrect}; reasoning {sound, incomplete, invalid, absent}; execution {accurate, slip, not-assessable}; confidence; familiarity; hint; time; exposure. Do not collapse these into a single ability score.

A rung is demonstrated provisionally by one sound independent response, corroborated only by a second independent changed example that probes the same claim. If a rung lacks a second observation, say insufficient corroboration. Partial/contradictory results require targeted checks. Higher-rung success can supply evidence for a prerequisite only when the written work makes that prerequisite visible.

References informing design

- ETS evidence-centered design: https://www.ets.org/research/policy_research_reports/publications/report/2011/imbu.html
- MAA competition domains: https://maa.org/student-programs/amc/
- MAA invitational competition overview: https://maa.org/maa-invitational-competitions/

These sources inform the design; they do not validate this question bank or its difficulty labels.